

Experiment 0: Software and Environment Setup

Title: Installation of required software and environment Setup

Objective: Familiarize students with tools and environments commonly used in ML.

Prerequisites :

- Basic knowledge of **Python programming**.
- Familiarity with using a command-line interface (CLI) or terminal.

Theory:

Install Python (with Anaconda for ease of package management).

Set up IDEs like Jupyter Notebook, Google Colab, or VS Code.

Install essential libraries:

- NumPy, pandas, Matplotlib, Seaborn (data handling and visualization).
- Scikit-learn, TensorFlow, PyTorch (ML frameworks).
- Statsmodels (for statistical analysis).

Setting Up Python Environment for ML

- Why Anaconda?
 - **Anaconda** is a popular distribution of Python and R for data science and machine learning. It comes pre-installed with many libraries and tools useful for ML.
- Setting up Python IDEs for ML
 - Jupyter Notebook: An open-source, web-based interactive computing environment that supports live code, visualizations, and documentation in notebooks. It's widely used for ML research and education.
 - Google Colab: A cloud-based IDE for running Jupyter notebooks that allows access to free GPU resources, useful for deep learning tasks.
 - Visual Studio Code (VS Code): A lightweight and powerful code editor with many extensions available for Python and ML development.

- **Essential Python Libraries for ML**

- **Scikit-learn:** A simple and efficient library for data mining and data analysis. It provides a wide range of tools for data preprocessing, model building, and evaluation.
- **Pandas:** A library for data manipulation and analysis, offering data structures like DataFrames.
- **NumPy:** Provides support for large, multi-dimensional arrays and matrices, along with mathematical functions to operate on them.
- **Matplotlib:** A plotting library for creating static, animated, and interactive visualizations.
- **TensorFlow & Keras:** Frameworks for deep learning, offering tools to build and train neural networks.

→ **Install Python with Anaconda (Recommended for ML)**

1. Download Anaconda:

- Visit the Anaconda Download Page and choose the appropriate version for your operating system (Windows/macOS/Linux).

2. Install Anaconda:

- Follow the installation instructions provided on the website.
- Ensure that you select the option to **add Anaconda to the system PATH** during installation.

3. Verify Installation:

- Open a terminal or Anaconda Prompt.
- Type the following command to check if Anaconda is installed correctly:
bash
`conda --version`
- You should see the version number of Anaconda.

(Youtube Link: <https://www.youtube.com/watch?v=M7Q8v9OggDU>)

→ **Set Up IDEs for Python and ML**

1. **Jupyter Notebook:**

- Jupyter Notebook comes pre-installed with Anaconda. If you're using Miniconda, install Jupyter Notebook via:
bash
`conda install jupyter`
- Launch Jupyter Notebook by typing the following command in your terminal:
bash
`jupyter notebook`
- This will open a web browser where you can create and manage notebooks.
(Youtube Link: https://youtu.be/pro5x_02FfU)

2. **Google Colab (Cloud-based):**

- Google Colab requires no installation.
- Simply visit Google Colab and sign in with your Google account.
- Create a new notebook and start coding with Python and ML libraries.
- **Tip:** Colab provides free access to GPUs for ML tasks.

3. **Visual Studio Code (VS Code):**

- Download and install [VS Code](#).
- After installation, open VS Code and go to the **Extensions** tab.
- Search for and install the **Python** extension.
- Install the **Jupyter** extension if you want to work with Jupyter Notebooks inside VS Code.

→ **Install Essential Libraries Using Conda or Pip**

1. **Install Libraries via Conda:**

bash
`conda install scikit-learn pandas numpy matplotlib tensorflow keras`

2. Install Libraries via Pip (if not using Anaconda):

bash

```
pip install scikit-learn pandas numpy matplotlib tensorflow keras
```

3. Check Installation:

Verify the installation of libraries by running a simple test:

python

```
import sklearn
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import tensorflow as tf
```

```
import keras
```

Conclusion:

By the end of this lab students will be comfortable installing and configuring Python and popular IDEs like Jupyter Notebook, Google Colab, and VS Code. They will also have learned how to install essential Python libraries like **Scikit-learn**, **Pandas**, **NumPy**, **Matplotlib**, **TensorFlow**, and **Keras**, which are widely used for machine learning tasks.

Practice Questions

- Exercise 1: Launch and Use Jupyter Notebook
- Exercise 2: Write and Run Python Code in Google Colab
- Exercise 3: Write and Run Python Code in VS Code

Viva Questions:

1. What is the purpose of using Anaconda for Python development in machine learning?
2. What is the command to check if Conda has been successfully installed on your system?
3. What are the steps to install Python and Jupyter Notebook using Anaconda?
4. What are the advantages of using Google Colab over Jupyter Notebook installed locally?
5. How would you install the necessary libraries like Scikit-learn, Pandas, and TensorFlow using Conda?
6. What are the key differences between Jupyter Notebooks and Google Colab?
7. How do you start a new Jupyter notebook session?
8. What are the key features of TensorFlow and Keras, and how are they related?

References

- Python Documentation: <https://docs.python.org>
- Anaconda Installation: <https://www.anaconda.com/products/individual>
- Jupyter Notebooks Documentation: <https://jupyter.org/>
- VS Code Setup for Python: <https://code.visualstudio.com/docs/python/python-tutorial>
- Google Colab Documentation: <https://colab.research.google.com>